

ACML101
Physical Chemistry Tutorial IV

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- Write down the Schrödinger equation for the following systems: (a) a particle of mass m in a cubical box of side a ; (b) a particle of mass m in a spherical box of radius a ; (c) a particle of mass m moving on the x -axis subjected to a force directed towards the origin, of magnitude proportional to the distance from the origin; (d) an electron moving in the presence of a nuclear charge $+Ze$; (e) two electrons moving in the presence of a fixed nucleus of charge $+Ze$.

Solution: (a) $-\frac{\hbar^2}{2m} \nabla^2 \psi(x, y, z) = E\psi(x, y, z)$

(b) $-\frac{\hbar^2}{2m} \nabla^2 \psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$

(c) You are given that $F = -kx$, where x is the displacement from equilibrium.

Given that $F = -\frac{dV}{dx}$, we find that the potential $V = -\int F dx = \frac{1}{2}kx^2$. So the

Schrödinger equation is $-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}kx^2\psi(x) = E\psi(x)$.

(d) $\left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r}\right) \psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$

(e)

$$\left(-\frac{\hbar^2}{2m_e} \nabla_1^2 - \frac{\hbar^2}{2m_e} \nabla_2^2 - \frac{Ze^2}{4\pi\epsilon_0 r_1} - \frac{Ze^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}}\right) \psi(r_1, \theta_1, \phi_1, r_2, \theta_2, \phi_2) = E\psi(r_1, \theta_1, \phi_1, r_2, \theta_2, \phi_2)$$

$$r_{12} = |r_1 - r_2|$$

2. When lithium is irradiated with light, one finds a stopping potential of 1.83 V for $\lambda = 3000\text{\AA}$ and 0.80 V for $\lambda = 4000\text{\AA}$. From the data, calculate (a) Planck's constant, (b) threshold frequency, and (c) work function of Li.

$$h\nu = \phi + KE$$

$$= \phi + \text{stopping potential} \times \text{electric charge}$$

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_0} + (\text{Stopping potential})_1$$

$$\frac{hc}{\lambda_2} = \frac{hc}{\lambda_0} + (\text{Stopping potential})_2$$

Two equation, two unknown $\rightarrow$ solve

h and λ_0

$$\text{Work function} = \frac{hc}{\lambda_0}$$

3. Identify which of the following functions are eigenfunctions of the operators d/dx and d^2/dx^2 : (a) e^{ikx} (b) $\cos kx$ (c) k (d) kx (e) e^{-ax^2} . Give the corresponding eigenvalue where appropriate.

$$(a) \quad \frac{d}{dx} \exp(ikx) = (ik) e^{ikx}$$

$\exp(ikx)$ is eigen-fn. of $\frac{d}{dx}$ and
eigen value is ik —

4. Write out the operator for $\hat{A}^2$ for $\hat{A} =$

(a) $\frac{d^2}{dx^2}$

(b) $\frac{d}{dx} + x$

(c) $\frac{d^2}{dx^2} - 2x \frac{d}{dx} + 1$

(b) $\hat{A} = \frac{d}{dx} + x$

$$\hat{A}^2 f(x) = \left(\frac{d}{dx} + x \right) \left[\left(\frac{d}{dx} + x \right) f(x) \right]$$

$$= \left(\frac{d}{dx} + x \right) \left(f'(x) + x f(x) \right)$$

$$= f''(x) + x f'(x) + f(x) + x f'(x) + x^2 f(x)$$

$$= f''(x) + 2x f'(x) + x^2 f(x) + f(x)$$

$$= \left(\frac{d^2}{dx^2} + 2x \frac{d}{dx} + x^2 + 1 \right) f(x)$$

$$\hat{A}^2 = \frac{d^2}{dx^2} + 2x \frac{d}{dx} + x^2 + 1$$

5. Evaluate the commutator $[\hat{A}, \hat{B}]$, where $\hat{A}$ and $\hat{B}$ are given below:

	$\hat{A}$	$\hat{B}$
(a)	$\frac{d}{dx} - x$	$\frac{d}{dx} + x$
(b)	$\frac{d^2}{dx^2} - x$	$\frac{d}{dx} + x^2$

(a)

$$\begin{aligned}
 \hat{A} \hat{B} f(x) &= \left(\frac{d}{dx} - x \right) \left(\frac{d}{dx} + x \right) f(x) \\
 &= \left(\frac{d}{dx} - x \right) \left(f'(x) + x f(x) \right) \\
 &= f''(x) + x f'(x) + f(x) - x f'(x) - x^2 f(x) \\
 &= f''(x) + (1 - x^2) f(x)
 \end{aligned}$$

$$\begin{aligned}
 \hat{B} \hat{A} f(x) &= \left(\frac{d}{dx} + x \right) \left(\frac{d^2}{dx^2} - x \right) f(x) \\
 &= \left(\frac{d}{dx} + x \right) \left(f''(x) - x f(x) \right) \\
 &= f''(x) - x f'(x) - f(x) - x f'(x) - x^2 f(x) \\
 &= f''(x) - (1 + x^2) f(x)
 \end{aligned}$$

$$\hat{A} \hat{B} \neq \hat{B} \hat{A}$$

$$\begin{aligned}
 [\hat{A}, \hat{B}] f(x) &= \cancel{f''(x) + (1 - x^2) f(x)} \\
 &\quad - \cancel{f''(x) + (1 + x^2) f(x)} \\
 &= 2f(x)
 \end{aligned}$$

$$[\hat{A}, \hat{B}] = 2$$

6. Two (unnormalized) excited state wavefunctions of the H-atom are (i) $\psi = \left(2 - \frac{r}{a_0}\right) \exp(-r/a_0)$ and (ii) $\psi = r \sin \theta \cos \phi \exp(-r/a_0)$. Normalize both functions to 1. Now show that these 2 functions are mutually orthogonal.

$$\psi_1 = N_1 \left(2 - \frac{r}{a_0}\right) \exp(-r/2a_0)$$

$$\psi_2 = N_2 r \sin \theta \cos \phi \exp(-r/2a_0)$$

$$\begin{aligned} \int \psi_1^2 d\tau &= N^2 \int_0^\infty \left(2 - \frac{r}{a_0}\right)^2 \times r^2 \exp(-r/a_0) \times r^2 dr \\ &\quad \times \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \\ &= 32 \pi a_0^3 N^2 = 1 \\ N_1 &= \left(\frac{1}{32 \pi a_0^3}\right)^{1/2} \end{aligned}$$

$$\text{similarly } N_2 = \frac{1}{32 \pi a_0^5}$$

Orthogonal

$$\int \psi_1 \psi_2 d\tau = 0$$

$$\begin{aligned} \text{Use } \int_0^\infty x^n \exp^{-ax} dx &= \frac{n!}{a^{n+1}} \\ \int_0^\pi \cos^n \theta \sin \theta d\theta &= - \int_1^{-1} \cos^n \theta d \cos \theta \\ &= + \int_{-1}^1 x^n dx \\ \int_0^{2\pi} \cos \phi d\phi &= 0 \end{aligned}$$

7. (a) Is $\psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x \exp[-\frac{1}{2}\alpha x^2]$ is an eigenfunction of the kinetic energy operator where $\alpha = \sqrt{k\mu}/\hbar^2$. (b) What are the average values of kinetic and potential energies for a quantum mechanical oscillator in this state.

~~Q7~~ $\psi_1(u) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\frac{1}{2}\alpha x^2}$ ~~and~~ $\alpha = \sqrt{\frac{k\mu}{\hbar^2}}$

(a) K.E. operator $\rightarrow -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \Rightarrow -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

see for yourself that $\psi_1(u)$ is not an eigenfn of the K.E. operator.

$$\begin{aligned} (b) \langle \text{Potential} \rangle &= \int \psi_1^*(u) V(u) \psi_1(u) du \\ &= \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \int_{-\infty}^{+\infty} x e^{-\frac{1}{2}\alpha x^2} \underbrace{\left(\frac{1}{2} k x^2\right)}_{V(u)} x e^{-\frac{1}{2}\alpha x^2} dx \\ &= \frac{1}{2} k \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \int_{-\infty}^{+\infty} x^4 e^{-\alpha x^2} dx \\ &= \underbrace{k \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \int_0^{\infty} x^4 e^{-\alpha x^2} dx} \end{aligned}$$

Use the standard integral

$$\int_0^{\infty} x^{2n} e^{-ax^2} dx = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2^{n+1} a^n} \sqrt{\frac{\pi}{a}}$$

$$\begin{aligned} \therefore \langle \text{Potential} \rangle &= \frac{1}{2} k \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \left(\sqrt{\frac{\pi}{\alpha}}\right) \frac{3}{4\alpha^2} \\ &= \frac{3k}{4\alpha} = \frac{3}{4} \hbar \sqrt{\frac{k}{\mu}} \end{aligned}$$

$$\begin{aligned}
\text{Also } \langle E_{\text{kinetic}} \rangle &= \int_{-\infty}^{+\infty} \psi_1^* \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \psi_1(x) dx \\
&= \left(-\frac{\hbar^2}{2m} \right) \left(\frac{4\alpha^3}{\pi} \right)^{1/2} \int_{-\infty}^{+\infty} x e^{-\alpha x^2/2} \left[\frac{d^2}{dx^2} (x e^{-\alpha x^2/2}) \right] dx \\
&= -\frac{\hbar^2}{2m} \left(\frac{4\alpha^3}{\pi} \right)^{1/2} \int_{-\infty}^{+\infty} (\alpha^2 x^4 - 3\alpha x^2) e^{-\alpha x^2} dx \\
&= -\frac{\hbar^2}{m} \left(\frac{4\alpha^3}{\pi} \right)^{1/2} \int_0^{\infty} (\alpha^2 x^4 - 3\alpha x^2) e^{-\alpha x^2} dx \\
&= \frac{3}{4} \frac{\hbar^2 \alpha}{m} = \underline{\underline{\frac{3}{4} \hbar \sqrt{\frac{\kappa}{m}}}}
\end{aligned}$$

Please note that

$$\langle P.E. \rangle = \langle K.E. \rangle$$

8. What is the probability of finding a particle in a 1-D box of length L : (a) in the right half of the box (b) in the central third of the box and (c) between $x = 0$ and $x = L/n$ when it is in the n -th state? Does this probability depend on the quantum number labeling the state of the particle?

Ans: 0.5, 0.61, $1/n$

(a) Right-half of the box.

$$\int_{L/2}^L \left(\frac{2}{L}\right) \sin^2 \frac{n\pi x}{L} dx$$

$$= \frac{2}{L} \int_{L/2}^L \sin^2 \frac{n\pi x}{L} dx$$

$$\left[\text{Using } \int \sin^2(by) dy = \frac{y}{2} - \frac{1}{4b} (\sin 2by) \right]$$

$$= \frac{2}{L} \left[\left[\frac{x}{2} \right]_{L/2}^L - \frac{1}{4 \cdot \frac{n\pi}{L}} \left[\sin \frac{2n\pi}{L} x \right]_{L/2}^L \right]$$

$$= \frac{2}{L} \left[\left(\frac{L}{2} - \frac{L}{4} \right) - \frac{L}{4n\pi} \left\{ \sin \frac{2n\pi}{L} \cdot L - \sin \frac{2n\pi}{L} \cdot \frac{L}{2} \right\} \right]$$

$$= \frac{2}{L} \left[\frac{L}{4} - \frac{L}{4n\pi} \left\{ \underset{\substack{\uparrow \\ 0}}{\sin 2n\pi} - \underset{\substack{\uparrow \\ 0}}{\sin n\pi} \right\} \right]$$

$$= \frac{2}{L} \cdot \frac{L}{4} = \frac{1}{2}$$

$$= \frac{1}{2} \quad \Leftarrow \text{ makes sense}$$

(b) central third of the box:

$$\begin{aligned} P &= \frac{2}{L} \int_{L/3}^{2L/3} \sin^2\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{2}{L} \left\{ \left[\frac{x}{2} \right]_{L/3}^{2L/3} - \frac{1}{4n\pi} \left[\sin \frac{2n\pi}{L} x \right]_{L/3}^{2L/3} \right\} \\ &= \frac{2}{L} \left\{ \left(\frac{2L}{6} - \frac{L}{6} \right) - \frac{L}{4n\pi} \left[\sin \frac{2n\pi}{L} \cdot \frac{2L}{3} - \sin \frac{2n\pi}{L} \cdot \frac{L}{3} \right] \right\} \\ &= \frac{2}{L} \left\{ \frac{L}{6} - \frac{L}{4n\pi} \left[\sin \frac{4n\pi}{3} - \sin \frac{2n\pi}{3} \right] \right\} \\ &= 2 \left[\frac{1}{6} - \frac{1}{4n\pi} \left(\sin \frac{4n\pi}{3} - \sin \frac{2n\pi}{3} \right) \right] \end{aligned}$$

$$\begin{aligned} \underline{n=1}; \quad P &= 2 \left[\frac{1}{6} - \frac{1}{4\pi} \left(\sin \frac{4\pi}{3} - \sin \frac{2\pi}{3} \right) \right] \\ &= 2 \left[\frac{1}{6} - \frac{1}{4\pi} \left[-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right] \right] \\ &= 2 \left[\frac{1}{6} + \frac{1}{4\pi} \sqrt{3} \right] \approx \underline{\underline{0.61}} \end{aligned}$$

Subsequently do it for other values of n .

$$(c) \quad \frac{2}{L} \int_0^{L/n} \sin^2 \left(\frac{n\pi x}{L} \right) dx$$

$$= \frac{2}{L} \left[\left\{ \frac{x}{2} \right\}_0^{L/n} - \frac{1}{4n\pi} \left\{ \sin \left(\frac{2n\pi}{L} \cdot x \right) \right\}_0^{L/n} \right]$$

$$= \frac{2}{L} \left[\frac{L}{2n} - \frac{1}{4n\pi} \left\{ \sin \frac{2n\pi}{L} \cdot \frac{L}{n} - 0 \right\} \right]$$

$$= 2 \left[\frac{1}{2n} - \frac{1}{4n\pi} \underbrace{\sin 2\pi}_0 \right]$$

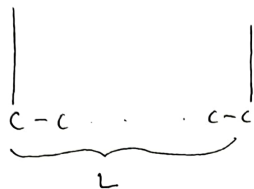
$$= \frac{1}{n}$$

Hence P does depend upon 'n'.

9. (i) To a crude first approximation, a π electron in linear polyene may be considered to be a particle in a one-dimensional box. The polyene in β - carotene contains 22 conjugated C atoms and the average internuclear distance is 140 pm. Each state upto $n = 11$ is occupied by two electrons. Calculate (a) the separation energy between the ground state and the first excited state in which one electron occupies the state with $n = 12$ and (b) The frequency of the radiation required to produce a transition between these two states.
- (ii) When β - carotene is oxidized, it breaks into half and forms two molecules of retinal (vitamin A) which is a precursor to the pigment in the retina responsible for vision. The conjugated system for retinal consists of 11 C atoms and one O atom. In the ground state of retinal, each level upto $n = 6$ is occupied by 2 electrons. Treating everything else to be similar repeat calculations for parts (a) and (b) of the previous problem keeping in mind that in this case the first excited state has one electron in the $n = 7$ state.

Ans: [$1.6 \times 10^{-19} J$; $2.41 \times 10^{14} s^{-1}$]

- (i) 22 conjugated carbon atoms and avg. internuclear distance
 $\approx 140 \text{ pm}$



$$\begin{aligned} L &= 21 (\text{bonds}) \times 140 \text{ pm} \\ &= 2940 \text{ pm} \\ &= 2940 \times 10^{-12} \text{ m} \end{aligned}$$

HOMO $\rightarrow$ corresponds to $n = 11$

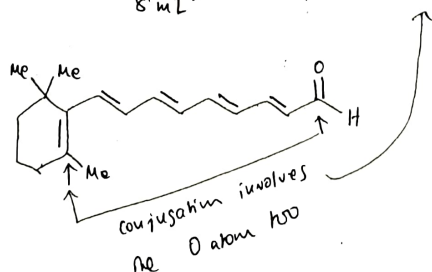
LUMO $\rightarrow$ " " $n = 12$

(a) $\Delta E = E_{12} - E_{11} = \frac{h^2}{8mL^2} (12^2 - 11^2) \approx 1.6 \times 10^{-19} \text{ J}$

$[m = 9.11 \times 10^{-31} \text{ kg}; L = 2940 \times 10^{-12} \text{ m}]$

(b) $\Delta E = h\nu \therefore \nu = 2.41 \times 10^{14} \text{ s}^{-1}$

(ii) $\Delta E = \frac{h^2 (7^2 - 6^2)}{8mL^2}; L = 11 \times 140 \text{ pm}$



$\Delta E = 3.3 \times 10^{-19} \text{ J}$

$\nu = 4.95 \times 10^{14} \text{ s}^{-1}$

Comment: Absorption spectrum shifts to ~~lower~~ lower frequency for a linear polyene as the number of conjugated atoms increases.

10. What are the most likely locations of a particle in a box of length L in the state $n = 3$.

Ans: $(L/6, L/2, 5L/6)$

$$\psi_3 = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{3\pi x}{L}\right)$$

$$P(x) \propto \psi_3^2 \propto \sin^2\left(\frac{3\pi x}{L}\right)$$

$$\frac{dP(x)}{dx} = 0 \quad \leftarrow \text{maxima and minima}$$

$$\frac{dP(x)}{dx} \propto 2 \sin\left(\frac{3\pi x}{L}\right) \cos\left(\frac{3\pi x}{L}\right) \propto \sin\left(\frac{6\pi x}{L}\right)$$

$$[2 \sin x \cos x = \sin 2x]$$

$$\sin\left(\frac{6\pi x}{L}\right) = 0 \quad \therefore \quad \frac{6\pi x}{L} = n\pi, \quad n = 0, 1, 2, \dots$$

$$\therefore \quad x = \frac{nL}{6} \quad n \leq 6$$

$n = 0, 2, 4$ and 6 correspond to minima in ψ_3

$$\text{example: } n=2, \quad \sin\left(\frac{6\pi x}{L}\right) = \sin\left(\frac{6\pi}{L} \cdot \frac{2L}{6}\right) = \sin(2\pi) = 0$$

here $n = 1, 3$ and 5 should correspond to the maxima that is

$$x = \frac{L}{6}, \quad \frac{3L}{6} \left(\frac{L}{2}\right), \quad \frac{5L}{6}$$

11. Show that the function $\exp(-\beta x^2)$ satisfies the Schrödinger equation for the quantum harmonic oscillator. What conditions does this place on β ? What is E ?

$$1. \quad - \frac{\hbar^2}{2\mu} \frac{d^2 \psi_n(u)}{du^2} + V(u) \psi_n(u) = E_n \psi_n(u)$$

↑
reduced mass

$$\therefore - \frac{\hbar^2}{2\mu} \frac{d^2}{du^2} [e^{-\beta u^2}] + \frac{1}{2} k u^2 (e^{-\beta u^2})$$

$$= - \frac{\hbar^2}{2\mu} [-2\beta e^{-\beta u^2}] + \frac{\hbar^2}{2\mu} [-4\beta^2 u^2 e^{-\beta u^2}] + \frac{1}{2} k u^2 e^{-\beta u^2}$$

$$= \frac{2\beta \hbar^2}{2\mu} e^{-\beta u^2} + \left[\frac{1}{2} k u^2 - \frac{4\beta^2 u^2 \hbar^2}{2\mu} \right] e^{-\beta u^2} \dots \textcircled{1}$$

$\underbrace{\hspace{10em}}_{=0}$
 (for $e^{-\beta u^2}$ to be an eigenfn, this factor has to go to zero)

$$\therefore \frac{1}{2} k = \frac{4\beta^2 \hbar^2}{2\mu}$$

$$\therefore \underline{\underline{\beta^2 = \frac{1}{4} \frac{k\mu}{\hbar^2}}}$$

Also from $\textcircled{1}$

$$E = \frac{\beta \hbar^2}{\mu} = \frac{\hbar^2}{\mu} \left(\frac{1}{4} \frac{k\mu}{\hbar^2} \right)^{1/2}$$

$$= \underline{\underline{\frac{\hbar}{2} \sqrt{\frac{k}{\mu}}}}$$